

Chapter 12

Transfer Functions and
Autoregressive Distributed Lag
Modeling

Chapter Outline

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TRANSFER FUNCTION-NOISE MODELS

12.1 The models that have been developed so far in this book have all been *univariate*, so that the current value of a time series depends, linearly or otherwise, only on past values of itself and, perhaps, a deterministic function of time. While univariate models are important in themselves, they also play a key role in providing a “baseline” to which *multivariate* models may be compared. We shall analyze several multivariate models over the next chapters, but our development begins with the simplest. This is the *single-input transfer function-noise* model, in which an *endogenous*, or *output*, variable y_t is related to a single *input*, or *exogenous*, variable x_t through the dynamic model¹

$$y_t = v(B)x_t + n_t \tag{12.1}$$

where the lag polynomial $v(B) = v_0 + v_1B + v_2B^2 + \dots$ allows x to influence y via a *distributed lag*: $v(B)$ is often referred to as the *transfer function* and the coefficients v_i as the *impulse response weights*.

12.2 It is assumed that both input and output variables are stationary, perhaps after appropriate transformation. The relationship between the two is not, however, deterministic—rather, it will be contaminated by noise captured by the stochastic process n_t , which will generally be serially correlated. A crucial assumption made in (12.1) is that x_t and n_t are independent, so that past x ’s influence future y ’s but not vice-versa, so ruling out *feedback* from y to x .

12.3 In general, $v(B)$ will be of infinite order and, hence, some restrictions must be placed on the transfer function before empirical modeling of (12.1)

becomes feasible. The typical way in which restrictions are imposed is analogous to the approximation of the linear filter representation of a univariate stochastic process by a ratio of low order polynomials in B , which leads to the familiar ARMA model (cf. §§3.25–3.27). More precisely, $v(B)$ may be written as the *rational distributed lag*

$$v(B) = \frac{\omega(B)B^b}{\delta(B)} \quad (12.2)$$

Here the numerator and denominator polynomials are defined as

$$\omega(B) = \omega_0 - \omega_1 B - \dots - \omega_s B^s$$

and

$$\delta(B) = 1 - \delta_1 B - \dots - \delta_r B^r$$

with the roots of $\delta(B)$ all assumed to be less than unity. The possibility that there may be a delay of b periods before x begins to influence y is allowed for by the factorization of the numerator in (12.2): if there is a contemporaneous relationship then $b = 0$.

12.4 The relationship between the impulse response weights v_i and the parameters $\omega_0, \dots, \omega_s, \delta_1, \dots, \delta_r$ and b can always be obtained by equating the coefficients of B^j in

$$\delta(B)v(B) = \omega(B)B^b$$

For example, if $r = 1$ and $s = 0$, so that

$$v(B) = \frac{\omega_0 B^b}{1 - \delta_1 B}$$

then

$$v_i = 0 \quad i < b$$

$$v_i = \omega_0 \quad i = b$$

$$v_i = \delta_1 v_{i-1} \quad i > b$$

Further examples are provided in Box and Jenkins (1970, Table 10.1) and Mills (1990, Table 13.1).

12.5 The noise process may be assumed to follow an ARMA (p, q) model:

$$n_t = \frac{\theta(B)}{\phi(B)} a_t$$

so that the combined transfer function-noise model can be written as

$$y_t = \frac{\omega(B)}{\delta(B)} x_{t-b} + \frac{\theta(B)}{\phi(B)} a_t \quad (12.3)$$

Box and Jenkins (1970, Chapter 11) develop an identification, estimation, and diagnostic checking procedure for single-input transfer function models of the form of (12.3). The identification stage uses the *cross-correlation function* between the output and input after they have both been transformed using the filter (i.e., ARMA model) that reduces x_t to white noise, which is known as *prewhitening*. Estimation and diagnostic checking use extensions of their univariate counterparts, although these are not necessarily straightforward. There have also been several other identification techniques proposed over the years, most notably based on prewhitening the input and output using individual filters (ARMA models).

12.6 If identifying a univariate ARMA model is often considered to be an “art form,” then identifying a transfer function in this way is even more so and, if there are multiple inputs, can become increasingly difficult, for the model is now:

$$y_t = \sum_{j=1}^M v_j(B)x_{j,t} + n_t = \sum_{j=1}^M \frac{\omega_j(B)B^{b_j}}{\delta_j(B)} x_{j,t} + \frac{\theta(B)}{\phi(B)} a_t \quad (12.4)$$

where

$$\omega_j(B) = \omega_{j,0} - \omega_{j,1}B - \dots - \omega_{j,s_j}B^{s_j}$$

and

$$\delta_j(B) = 1 - \delta_{j,1}B - \dots - \delta_{j,r_j}B^{r_j}$$

The simplest way to proceed is to use the Box–Jenkins approach in a “piecemeal” fashion, identifying a set of single-input transfer functions between y and x_1 , y and x_2 , etc., and then combining them to identify the noise model, after which estimation and diagnostic checking can be attempted.

12.7 The problem here is that the sample cross-correlation functions between each prewhitened input and the correspondingly filtered output may be misleading if the x ’s are intercorrelated. In mitigation, if these input series have been differenced to induce stationarity then it is well known that such differencing will reduce intercorrelations between the series. It may then be the case that the stationary input series are only weakly related and, analogous to the situation of independent regressors in multiple regression analysis, this piecemeal approach to identification may often work quite well, particularly as joint estimation of the combined model and judicious use of diagnostic checking procedures may then enable an adequately specified model to be obtained.

AUTOREGRESSIVE DISTRIBUTED LAG MODELS

12.8 Nevertheless, it would clearly be useful if an automatic model selection procedure could be developed. This has not been done for the multiple

input model (12.4), but if a restricted form of it is specified then such a procedure becomes feasible. This restricted form is known as the *autoregressive distributed lag*, or ARDL, model and is obtained by placing the following restrictions on (12.4):

$$\delta_1(B) = \dots = \delta_M(B) = \phi(B) \quad \theta(B) = 1$$

so that the model is, on defining $\beta_j(B) = \omega_j(B)B^{b_j}$ and including an intercept,

$$\phi(B)y_t = \beta_0 + \sum_{j=1}^M \beta_j(B)x_{j,t} + a_t \quad (12.5)$$

This is known as the ARDL($p, s_1, \dots, s_M$) model and restricts all the autoregressive lag polynomials to be the same and excludes a moving average noise component, although this exclusion is not essential. These restrictions reduce the noise component to white noise through constraining the dynamics and enables (12.5) to be estimated by OLS, so that on selecting a maximum lag order of, say, m , goodness-of-fit statistics, such as information criteria, can be used to select the appropriate specification.

12.9 The ARDL representation (12.5) may be recast in a potentially useful way. Recalling the development of §8.4, each input polynomial may be decomposed as

$$\beta_j(B) = \beta_j(1) + \nabla \tilde{\beta}_j(B)$$

where

$$\tilde{\beta}_j(B) = \tilde{\beta}_{j,0} + \tilde{\beta}_{j,1}B + \tilde{\beta}_{j,2}B^2 + \dots + \tilde{\beta}_{j,s_j-1}B^{s_j-1}$$

with

$$\tilde{\beta}_{j,i} = - \sum_{l=i+1}^{s_j} \beta_{j,l}$$

Consequently, (12.5) can be written as

$$y_t = \beta_0 + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^M \beta_j(1)x_{j,t} + \sum_{j=1}^M \tilde{\beta}_j(B) \nabla x_{j,t} + a_t \quad (12.6)$$

Solving for y_t then yields

$$y_t = \theta_0 + \sum_{j=1}^M \theta_j x_{j,t} + \sum_{j=1}^M \tilde{\theta}_j(B) \nabla x_{j,t} + \varepsilon_t \quad (12.7)$$

in which

$$\theta_0 = \phi^{-1}(1)\beta_0$$

$$\theta_j = \phi^{-1}(1)\beta_j(1)$$

$$\tilde{\theta}_j(B) = \phi^{-1}(B)(\tilde{\beta}_j(B) - \tilde{\phi}(B)\phi^{-1}(1)\beta_j(1)) \quad j = 1, \dots, M$$

$$\varepsilon_t = \phi^{-1}(B)a_t$$

where

$$\tilde{\phi}(B) = \nabla^{-1}(\phi(B) - \phi(1))$$

has been used.

The representation (12.7) separates out the *long-run* relationships between the output and the inputs from short-run effects, but is not amenable to direct estimation. The estimate of the long-run relationship between y and x_j may be obtained from (12.6) and is given by

$$\hat{\theta}_j = \frac{\hat{\beta}_j(1)}{1 - \sum_{i=1}^p \hat{\phi}_i} = \frac{\hat{\beta}_{j,0} + \dots + \hat{\beta}_{j,s_j}}{1 - \sum_{i=1}^p \hat{\phi}_i}$$

and an accompanying standard error may be computed accordingly.

Various other re-expressions of the ARDL model (12.5) are available and are discussed in detail in Banerjee et al. (1993, Chapter 2). One such transformation, known as the *error-correction* model, will be introduced and used extensively in Chapter 14, Error Correction, Spurious Regressions, and Cointegration.

EXAMPLE 12.1 An ARDL Model for UK Interest Rates

In Example 11.3 an ESTAR model was estimated for the monthly change in the United Kingdom long interest rate, which produced a modest improvement in fit over the linear AR(2) specification, estimated to be (with $DR20$ now denoting the change in the long rate):

$$DR20_t = \frac{0.304}{(0.035)} DR20_{t-1} - \frac{0.126}{(0.035)} DR20_{t-2} + \hat{a}_t \quad \hat{\sigma}_a = 0.279$$

We now fit an ARDL model with the change in the short interest rate, DRS , included as a single input. Setting the maximum lag order at $m=4$, Fig. 12.1 shows the AIC values for the 20 possible ARDL models, from which an ARDL (2,1) is selected and which is estimated to be

$$DR20_t = \frac{0.227}{(0.036)} DR20_{t-1} - \frac{0.109}{(0.036)} DR20_{t-2} + \frac{0.297}{(0.021)} DRS_t - \frac{0.073}{(0.023)} DRS_{t-1} + \hat{a}_t$$

$$\hat{\sigma}_a = 0.249$$

The equivalent form of (12.6) is

$$DR20_t = \frac{0.227}{(0.036)} DR20_{t-1} - \frac{0.109}{(0.036)} DR20_{t-2} + \frac{0.224}{(0.028)} DRS_t + \frac{0.075}{(0.023)} \nabla DRS_t + \hat{a}_t$$

and the long-run parameter contained in (12.7) is then calculated to be

$$\hat{\theta} = \frac{0.224}{1 - 0.227 + 0.109} = \frac{0.254}{(0.028)}$$

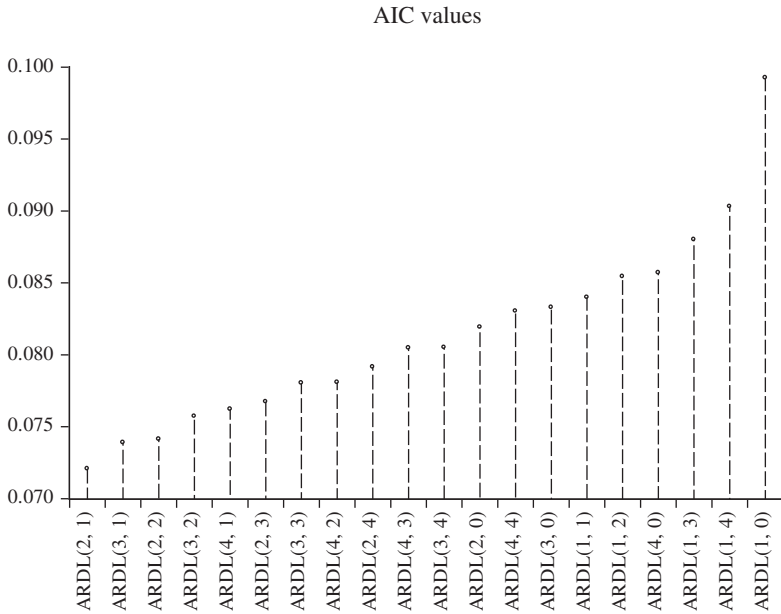


FIGURE 12.1 AIC values for ARDL models of short and long interest rates. *AIC*, Akaike Information criterion; *ARDL*, Autoregressive distributed lag.

The significance of both the contemporaneous and one-month lagged change in the short rate produces an 11% reduction in the residual standard error over the univariate model and shows that knowledge of changes in the short interest rate help to predict changes in the long interest rate.

EXAMPLE 12.2 ARDL Modeling of Global Temperatures

Fig. 12.2 shows global temperature and four possible “forcings”—total radiative forcing due to both anthropogenic and natural factors (excluding volcanic eruptions), radiative forcing due to volcanic stratospheric aerosols, and scaled indices of the Southern Oscillation and the Atlantic Multidecadal Oscillation—annually for the period of 1866–2015. From this figure, global temperature and total radiative forcing are clearly nonstationary, as would be expected, and unit root tests confirm they are both $I(1)$. Volcanic forcing and the two oceanic-atmospheric phenomena are, however, clearly stationary, as is also confirmed by unit root tests.

The ARDL model, (12.8), containing only stationary variables, was therefore considered, using obvious names for the variables to aid interpretation:

$$\begin{aligned} \nabla \text{TEMP}_t = & \beta_0 + \sum_{i=1}^m \phi_i \nabla \text{TEMP}_{t-i} + \sum_{i=0}^m \beta_{1,i} \nabla \text{TRF}_{t-i} + \sum_{i=0}^m \beta_{2,i} \text{VOLC}_{t-i} \\ & + \sum_{i=0}^m \beta_{3,i} \text{SO}_{t-i} + \sum_{i=0}^m \beta_{4,i} \text{AMO}_{t-i} + a_t \end{aligned} \quad (12.8)$$

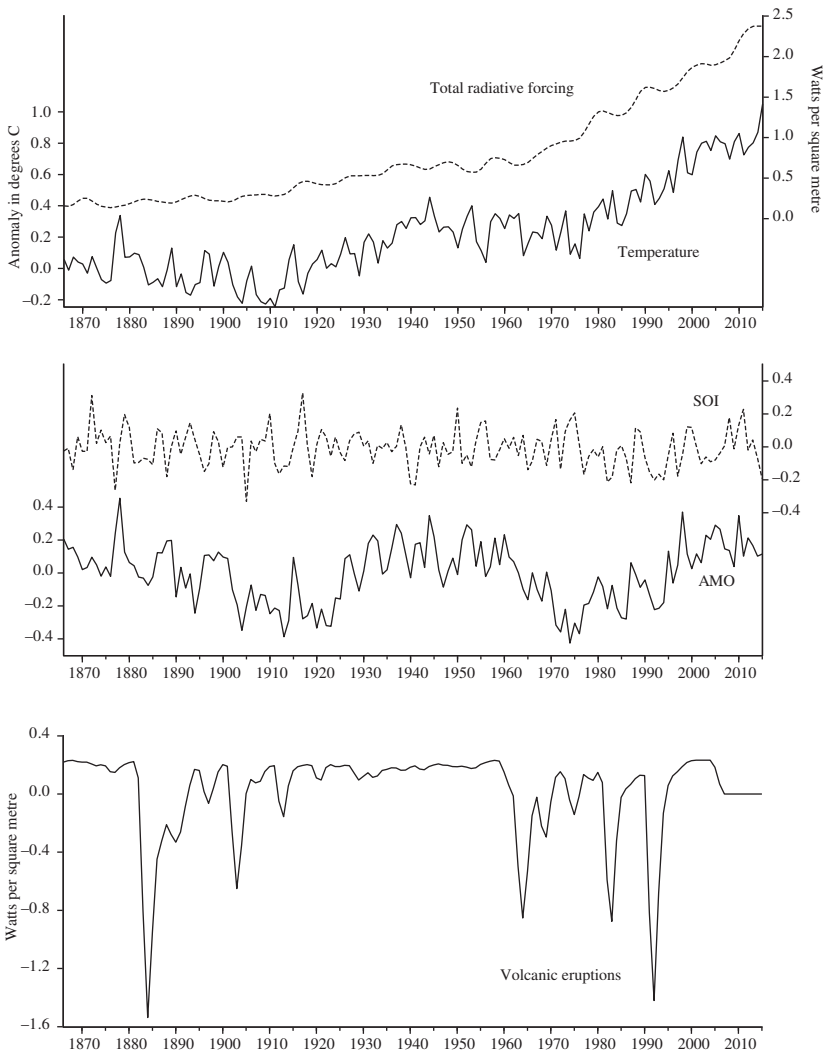


FIGURE 12.2 Climate data, annual 1866–2015. Top panel: temperature and total radiative forcing (excluding from volcanic eruptions); middle panel: Southern Oscillation index (SOI) and Atlantic Multidecadal Oscillation (AMO); bottom panel: radiative forcing from volcanic eruptions. McDermott, G.R., 2017. *Sceptic priors and climate policy. Working paper.*; data downloaded from <<https://github.com/grantrmcdermott/sceptic-priors>>.

All the forcing variables may be considered to be exogenous and, on setting $m = 6$, an ARDL(3,2,0,1,4) model was selected, whose parameter estimates are shown in column (1) of Table 12.1. Various residual checks confirm the adequacy of this specification. Several of the parameter estimates are insignificant

TABLE 12.1 ARDL(3,2,0,1,4) Estimates of (12.8)

	Unrestricted ARDL (1)	Restricted (2)	Restricted (3)
$\hat{\beta}_0$	− 0.002 (0.007)	—	—
$\hat{\phi}_1$	− 0.519 (0.079)	− 0.518 (0.071)	− 0.520 (0.070)
$\hat{\phi}_2$	− 0.350 (0.078)	− 0.369 (0.066)	− 0.370 (0.066)
$\hat{\phi}_3$	− 0.333 (0.074)	− 0.325 (0.067)	− 0.324 (0.067)
$\hat{\beta}_{1,0}$	0.622 (0.483)	—	—
$\hat{\beta}_{1,1}$	− 1.055 (0.761)	—	—
$\hat{\beta}_{1,2}$	1.160 (0.487)	0.560 (0.169)	0.561 (0.168)
$\hat{\beta}_{2,0}$	0.089 (0.022)	0.086 (0.021)	0.085 (0.021)
$\hat{\beta}_{3,0}$	− 0.336 (0.058)	− 0.342 (0.056)	− 0.343 (0.056)
$\hat{\beta}_{3,1}$	− 0.105 (0.066)	− 0.107 (0.063)	− 0.107 (0.063)
$\hat{\beta}_{4,0}$	0.379 (0.051)	0.382 (0.049)	0.385 (0.046)
$\hat{\beta}_{4,1}$	− 0.239 (0.067)	− 0.243 (0.058)	− 0.241 (0.057)
$\hat{\beta}_{4,2}$	− 0.017 (0.072)	—	—
$\hat{\beta}_{4,3}$	0.028 (0.069)	—	—
$\hat{\beta}_{4,4}$	− 0.154 (0.061)	− 0.147 (0.047)	− 0.144 (—)
$\hat{\sigma}_a$	0.0696	0.0689	0.0687

ARDL, Autoregressive distributed lag.

and were, therefore, set to zero, with the restricted model estimates being shown in column (2). Written in the form of (12.6), this model is

$$\begin{aligned}
 \nabla TEMP_t = & -\frac{0.518}{(0.071)} \nabla TEMP_{t-1} - \frac{0.369}{(0.066)} \nabla TEMP_{t-2} - \frac{0.325}{(0.067)} \nabla TEMP_{t-3} \\
 & + \frac{0.560}{(0.169)} \nabla TRF_{t-2} + \frac{0.086}{(0.021)} VOLC_t - \frac{0.449}{(0.082)} SO_t + \frac{0.107}{(0.063)} \nabla SO_t \\
 & - \frac{0.007}{(0.039)} AMO_t + \frac{0.243}{(0.058)} \nabla AMO_t + \frac{0.147}{(0.047)} \nabla^4 AMO_t + \hat{a}_t
 \end{aligned}
 \tag{12.9}$$

All the estimates are highly significant except for $\beta_4(1)$. Column (3) of Table 12.1 reports the estimates of the model when this coefficient is set to zero. With this restriction imposed, the model can be written in terms of the levels of each variable as

TABLE 12.2 ARDL(4,0,1,1) Estimates of (12.11)

	Unrestricted ARDL (1)	Restricted (2)
$\hat{\beta}_0$	− 0.046 (0.012)	− 0.046 (0.011)
$\hat{\phi}_1$	0.310 (0.080)	0.309 (0.078)
$\hat{\phi}_2$	0.083 (0.057)	0.094 (0.060)
$\hat{\phi}_3$	0.027 (0.063)	—
$\hat{\phi}_4$	0.133 (0.055)	0.144 (0.048)
$\hat{\beta}_{1,0}$	0.189 (0.483)	0.191 (0.032)
$\hat{\beta}_{2,0}$	0.050 (0.029)	0.048 (0.011)
$\hat{\beta}_{2,1}$	0.048 (0.029)	0.048 (—)
$\hat{\beta}_{3,0}$	− 0.307 (0.054)	− 0.305 (0.053)
$\hat{\beta}_{3,1}$	− 0.117 (0.061)	− 0.114 (0.060)
$\hat{\beta}_{4,0}$	0.390 (0.047)	0.393 (0.046)
$\hat{\beta}_{4,1}$	− 0.189 (0.012)	− 0.191 (0.056)
$\hat{\sigma}_a$	0.0659	0.0655

ARDL, Autoregressive distributed lag.

$$\begin{aligned}
 TEMP_t = & -0.520TEMP_{t-1} - 0.370TEMP_{t-2} - 0.324TEMP_{t-3} + 0.561TRF_{t-2} \\
 & + 0.085 \sum_{i=0}^{\infty} VOLC_{t-i} - 0.343SO_t - 0.450 \sum_{i=1}^{\infty} SO_{t-i} \\
 & + 0.385AMO_t + 0.144 \sum_{i=1}^3 AMO_{t-i} + \text{constant}
 \end{aligned}
 \tag{12.10}$$

The level of total radiative forcing, therefore, has a positive effect on temperatures, while it is the cumulative level of volcanic eruptions and the SO that effect temperatures, positively for the former and negatively for the latter. Because the long-run effect of the AMO on temperature change is zero, this index has only a transitory positive effect on temperature levels.

Of especial interest is the long-run coefficient on total radiative forcing, estimated to be

$$\hat{\theta}_1 = \frac{0.561}{1 + 0.520 + 0.370 + 0.324} = \frac{0.253}{(0.075)}$$

which has an accompanying 95% confidence interval of $0.1 < \theta_1 < 0.4$. The *transient climate response* (TCR) is defined as the warming produced after a doubling of CO₂ if CO₂ increases at a rate of 1% per annum, i.e., after 70 years. It may be calculated as $TCR = \hat{\theta}_1 \times F_{2\times}$, where $F_{2\times}$ is the change in forcing that results from such a doubling of CO₂. A value of $F_{2\times} = 3.71$ is the Intergovernmental Panel on Climate Change's (IPCC) best estimate of this parameter. Thus, TCR is

estimated to be 0.94°C with a 95% confidence interval for the TCR of $0.4 < \text{TCR} < 1.5$. This is very much at the lower end of the range that the IPCC thinks is “most likely,” this being 1.0°C – 2.5°C .

An alternative approach is to ignore the “ $I(1)$ -ness” of temperatures and total radiative forcing and to consider the “levels” model:

$$\begin{aligned} \text{TEMP}_t = & \beta_0 + \sum_{i=1}^m \phi_i \text{TEMP}_{t-i} + \sum_{i=0}^m \beta_{1,i} \text{TRF}_{t-i} + \sum_{i=0}^m \beta_{2,i} \text{VOLC}_{t-i} \\ & + \sum_{i=0}^m \beta_{3,i} \text{SO}_{t-i} + \sum_{i=0}^m \beta_{4,i} \text{AMO}_{t-i} + a_t \end{aligned} \quad (12.11)$$

Again setting $m = 6$, an $\text{ARDL}(4,0,1,1,1)$ model was selected, whose estimates are shown in column (1) of Table 12.2. On noting that the coefficient estimates on VOLC_t and VOLC_{t-1} are almost identical, the restriction $\beta_{20} - \beta_{21} = 0$ was imposed, along with omitting the insignificant third-order lag on temperatures, and this restricted model is shown in column (2) of Table 12.2. Writing it in the form of (12.6), this model is

$$\begin{aligned} \text{TEMP}_t = & -\frac{0.046}{(0.011)} + \frac{0.309}{(0.078)} \text{TEMP}_{t-1} + \frac{0.094}{(0.060)} \text{TEMP}_{t-2} + \frac{0.144}{(0.048)} \text{TEMP}_{t-4} \\ & + \frac{0.191}{(0.032)} \text{TRF}_t + \frac{0.096}{(0.022)} \text{VOLC}_t - \frac{0.048}{(-)} \nabla \text{VOLC}_t \\ & - \frac{0.419}{(0.078)} \text{SO}_t + \frac{0.114}{(0.060)} \nabla \text{SO}_t + \frac{0.202}{(0.049)} \text{AMO}_t + \frac{0.191}{(0.056)} \nabla \text{AMO}_t + \hat{a}_t \end{aligned}$$

This provides a slightly better fit than (12.9) and continues to pass all diagnostic checks. The long-run coefficient of total radiative forcing is now:

$$\hat{\theta}_1 = \frac{0.191}{1.0309 - 0.094 - 0.144} = \frac{0.422}{(0.021)}$$

and this gives a 95% confidence interval for the TCR of $1.53^{\circ}\text{C} < \text{TCR} < 1.61^{\circ}\text{C}$. This is rather higher and much more precise than that obtained from (12.9) and lies in the central region of the IPCC's range of likely values of the TCR.

ENDNOTES

1. Other terms that are used are *dependent* for y_t and *independent* or *pre-determined* for x_t .